GPIO ASSIGNMENTS — v2.0 FINAL

GPIO	Define / Function
16	LED_RED
26	LED_GREEN
32	LED_BLUE / MEMS_BCLK ⚠ shared
13	BTN_NEXT (INPUT_PULLUP)
14	BTN_ENT (INPUT_PULLUP)
0	BOOT button front ⚠
5	TFT_CS ⚠ boot sensitive
17	TFT_DC
12	TFT_BL BC847 ⚠ init LOW first
23	TFT_MOSI (VSPI shared)
18	TFT_SCLK (VSPI shared)
19	TFT_MISO (VSPI shared)
33	TOUCH_CS (XPT2046)
15	SD_CS ⚠ 10k pull-up fitted
21	I2C_SDA
22	I2C_SCL
4	ONE_WIRE (DS18B20)
39	LDR (ADC1 safe)
36	MIC_ANALOG (ADC1 safe)
35	MEMS_WS (input-only OK with I2S)
34	MEMS_SD (input-only)
25	DAC_AUDIO (DAC1)
27	PIEZO (LEDC ch 1)
1/3	UART TX/RX debug header

⚠ GPIO 6–11 reserved (flash) — NEVER USE
ADC2 disabled when WiFi active. Use ADC1 (GPIO 32–39) only.

platformio.ini BUILD FLAGS

```
[env:e4_educator_v2]
platform    = espressif32@6.6.0  ← ALWAYS PIN
board_build.partitions = min_spiffs.csv
board_build.filesystem = littlefs
build_flags =
  -DFW_VERSION=\"1.0.0\"  -DFW_DATE=\"2026-05-01\"
  -DLED_RED=16  -DLED_GREEN=26  -DLED_BLUE=32
  -DBTN_NEXT=13  -DBTN_ENT=14
  -DTFT_CS=5 -DTFT_DC=17 -DTFT_BL=12
  -DTFT_MOSI=23 -DTFT_SCLK=18 -DTFT_MISO=19
  -DTOUCH_CS=33  -DSD_CS=15
  -DI2C_SDA=21  -DI2C_SCL=22
  -DONE_WIRE=4 -DLDR=39 -DMIC_ANALOG=36
  -DMEMS_BCLK=32 -DMEMS_WS=35 -DMEMS_SD=34
  -DDAC_AUDIO=25  -DPIEZO=27
  -DUSER_SETUP_LOADED=1 -DILI9341_DRIVER=1
  -DTFT_WIDTH=320 -DTFT_HEIGHT=240 -DTFT_RST=-1
  -DSMOOTH_FONT=1 -DSPI_FREQUENCY=27000000
  -DSPI_TOUCH_FREQUENCY=2500000
  -DELEGANTOTA_USE_ASYNC_WEBSERVER=1
  -DDEFAULT_SSID=\"lab_wifi\"
  -DDEFAULT_BROKER=\"192.168.1.79\"
```

CRITICAL RULES

Rule	Detail
TFT first, SD second	Always tft.init() before SD.begin() — SPI bus config required
GPIO 12 LOW first	digitalWrite(TFT_BL,LOW) before ledcSetup() — boot voltage
Wire.begin() explicit	Wire.begin(I2C_SDA,I2C_SCL) before ANY I2C library
NVS keys ≤ 15 chars	Silent truncation above 15. Collisions cause wrong values. Count every key.
mqtt.loop() every pass	Without this, incoming messages lost and keepalive fails
ElegantOTA.loop()	OTA stalls mid-transfer without this every loop() pass
ADC1 only + WiFi	WiFi uses ADC2. Only GPIO 32-39 safe for analog when WiFi active.
onReceive() minimal	ESP-NOW callback runs in WiFi task. Copy + flag only. Process in loop().

TIER 1 — PL

T	Topic
T01	Platfo E4
T02	GPIO build_
T03	Button debou
T04	I2C fundar
T05	OLED
T06	One-v analog
T07	Audio

TIER 2 — CO

T	Topic
T08	TFT & pipelin
T09	Touch
T10	SD ca I/O
T11	NVS p
T12	WiFi &
T13	LittleF
T14	OTA u
T15	ESP-N
T16	I2S &

TOOLKIT FI

File
Button.h
TonePlaye
TouchZone
Settings.
SensorPac

```
// Sprite
spr.fillSp
spr.setTe
spr.setCu
spr.pushSp
// millis(
if (now -
  lastXxx
```

STORAGE FI

NVS
LittleFS
SD card

MarkPi3 — Raspberry Pi 3B

IP	192.168.1.79
MQTT	port 1883
Node-RED	:1880 ui→:1880/ui
SSID	lab_wifi
NTP	GMT+46800 (UTC+13)

Pinned library versions

espressif32	@6.6.0
TFT_eSPI	@2.5.43
PubSubClient	@2.8
ArduinoJson	@6.21.3
arduinoFFT	@1.6.2
ElegantOTA	@2.2.9

MQTT topic structure

e4/sensor/temperature	E4→broker retained
e4/sensor/humidity	E4→broker retained
e4/status	online/offline LWT
e4/firmware	FW_VERSION retained
e4/cmd/brightness	broker→E4 command

I2C addresses · RGB565 colo

0x38	AHT20 (fixed)
0x3C	SSD1306 OLED
0x76	BMP280/BME280

COL_BG 0x0000	Black backg
COL_HEADER 0x0319	Dark brand b
COL_OK 0x07E0	Green — no

e4/node/N/temp	ESP-NOW bridge

COL_WARN 0xFFE0	Yellow — ca
COL_ERR 0xF800	Red — alert
COL_ACCENT 0xFD20	Orange — h